

DAA Assignment – Set 4

Question 1 – Dynamic Programming (Optimal Binary Search Tree)

You are given n ordered keys $k_1, k_2, \dots, k_n$ with their successful search probabilities $p_1, p_2, \dots, p_n$ and unsuccessful search probabilities $q_0, q_1, \dots, q_n$.

Design a dynamic programming algorithm to construct an Optimal Binary Search Tree (OBST) that minimizes the expected search cost.

Tasks

1. Derive the recurrence relation for the expected search cost.
2. Implement the dynamic programming algorithm.
3. Construct the optimal BST using the computed tables.
4. Analyze the time and space complexity.
5. Experimentally evaluate the algorithm for different values of n .

Question 2 – Greedy Method (Huffman Coding)

You are given a set of characters and their frequencies.

Design a greedy algorithm to construct the Huffman code for the characters.

Tasks

1. Design the algorithm using an appropriate data structure.
2. Implement the Huffman coding algorithm.
3. Generate the prefix codes for the characters.
4. Compute the total number of bits required for encoding.
5. Analyze the time complexity.

Question 3 – Divide and Conquer (Karatsuba Multiplication)

Given two large integers represented as strings, design a divide-and-conquer algorithm to multiply them using the Karatsuba multiplication technique.

Tasks

1. Design the Karatsuba multiplication algorithm.
2. Implement the algorithm.
3. Derive the recurrence relation for the running time.
4. Solve the recurrence.
5. Compare its performance with the standard grade-school multiplication algorithm.